Homework #10

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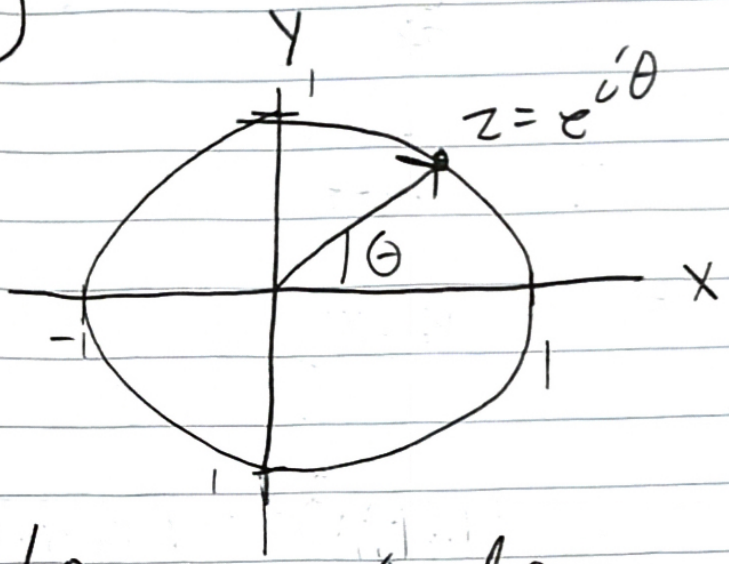
Prob 1.) $* \int_0^{2\pi} \frac{d\theta}{1 - 2t\cos\theta + t^2}$

$(|t| < 1)$



$* \; z = e^{i\theta}$

$* \; dz = ie^{i\theta}d\theta = iz\,d\theta$

$* \; \cos\theta = \frac{e^{i\theta} + e^{-i\theta}}{2} = \frac{z + 1/z}{2} = \frac{z + z^{-1}}{2}$

$$I = \frac{1}{i}\oint \frac{(1/z)\,dz}{1 - 2t\left(\frac{z+z^{-1}}{2}\right) + t^2}$$

$$I = \frac{1}{i}\oint \frac{(1/z)}{k - 2l\left(\frac{z+z^{-1}}{2}\right)}$$

Use simpler constants:

$* \; k = 1 + t^2$

$* \; l = t$

$$(1+t^2)^2 = 1+2t^2+t^4$$

$$I = \frac{1}{i}\oint \frac{dz}{kz - lz^2 - l}$$

$$I = \frac{-1}{i}\oint \frac{dz}{lz^2 - kz + l}$$

$$z = \frac{k \pm \sqrt{k^2 - 4(l)l}}{2l}$$

$$z = \frac{k \pm \sqrt{k^2 - 4l^2}}{2l}$$

$$z = \frac{(1+t^2) \pm \sqrt{1+2t^2+t^4 - 4t^2}}{2t}$$

$$z = \frac{(1+t^2) \pm \sqrt{t^4 + 2t^2 + 1}}{2t}$$

$$z = \frac{(1+t^2) \pm \sqrt{(1-t^2)^2}}{2t}$$

$$z = \frac{(1+t^2) \pm (1-t^2)}{2t}$$

$$z_1 = \frac{1+t^2-1+t^2}{2t} = \frac{2t^2}{2t} = t$$

$$z_2 = \frac{1+t^2+1-t^2}{2t} = \frac{2}{2t} = \frac{1}{t}$$

factors:

- $f_1 = (z-t)$
- $f_2 = \left(z-\frac{1}{t}\right)$

$$I = \frac{-1}{it}\oint \frac{dz}{(z-t)\left(z-\frac{1}{t}\right)}$$

* $|t|<1$ t is inside contour.

* $1/t$ is outside contour

$$f(z) = \frac{1}{(z-t)(z-1/t)}$$

$$R(t) = \lim_{z\to t}\left(z-t\right)\frac{1}{(z-t)(z-1/t)}$$

$$= \lim_{z\to t}\frac{1}{(z-1/t)} = \frac{1}{t-1/t}$$

$$R(t) = \frac{1}{t - \frac{1}{t}}$$

$$= \frac{1}{\frac{t^2}{t} - \frac{1}{t}}$$

$$= \frac{1}{\frac{t^2-1}{t}} \quad , \quad = \frac{t}{t^2-1}$$

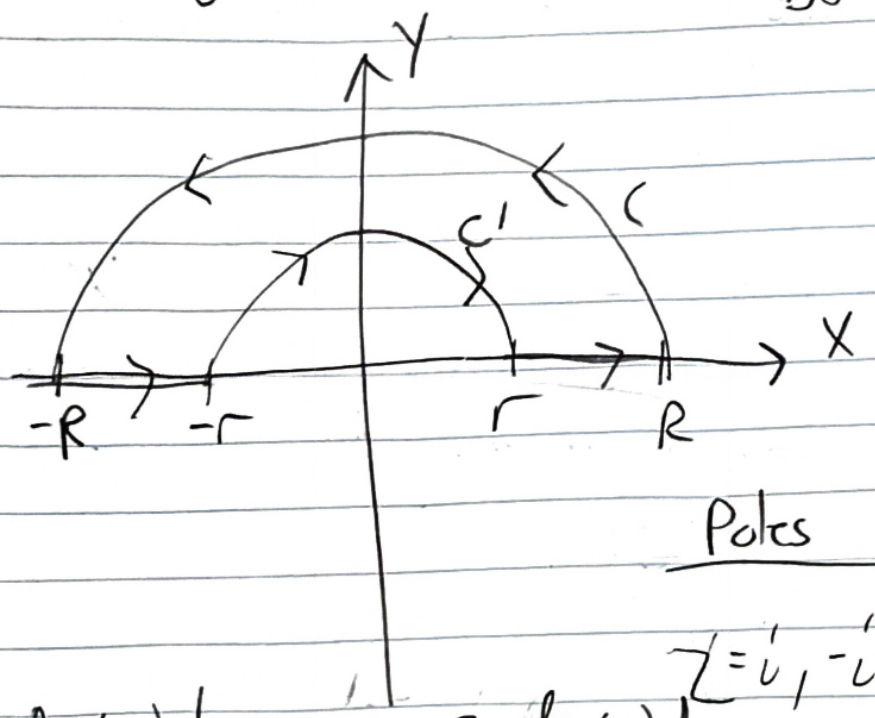
$$I = \frac{-1}{it} \oint \frac{dz}{(z-t)\left(z-\frac{1}{t}\right)} = 2\pi i \left(\frac{-1}{it}\right)\left(\frac{t}{t^2-1}\right)$$

$$\boxed{I = \frac{-2\pi}{t^2-1}}$$

Prob 2.)

- $\int_0^\infty \frac{\ln(x)\,dx}{1+x^2}$

- $I = \int_0^\infty \frac{\ln(x)\,dx}{1+x^2} = \frac{1}{2}\int_{-\infty}^\infty \frac{\ln(x)\,dx}{1+x^2}$



Poles:

$z = i, -i, 0$

- $\int_{-\infty}^\infty \frac{\ln(x)\,dx}{1+x^2} \to \oint \frac{\ln(z)\,dz}{1+z^2}$

- $f(z) = \frac{\ln(z)}{1+z^2}$ $\qquad R(i) = \lim_{z\to i} \frac{(z-i)\ln(z)}{(z+i)(z-i)}$

$$R(i) = \frac{i\pi/2}{2i} = \frac{\pi}{4}$$

$$\oint_C \frac{\ln(z)}{1+z^2}\,dz = \frac{-\pi^2}{2}\,i$$

$z = re^{i\theta}$

$dz = rie^{i\theta}\,d\theta$

$$\oint_C \frac{\ln(z)}{1+z^2}\,dz = \int_r^R \frac{\ln(x)}{1+x^2}\,dx + \oint_C \frac{\ln(re^{i\theta})\,ire^{i\theta}\,d\theta}{1+(re^{i\theta})^2} - \int_R^r \frac{\ln(xe^{i\pi})}{1+(xe^{i\pi})^2}\,e^{i\pi}\,dx$$

$r \to 0$

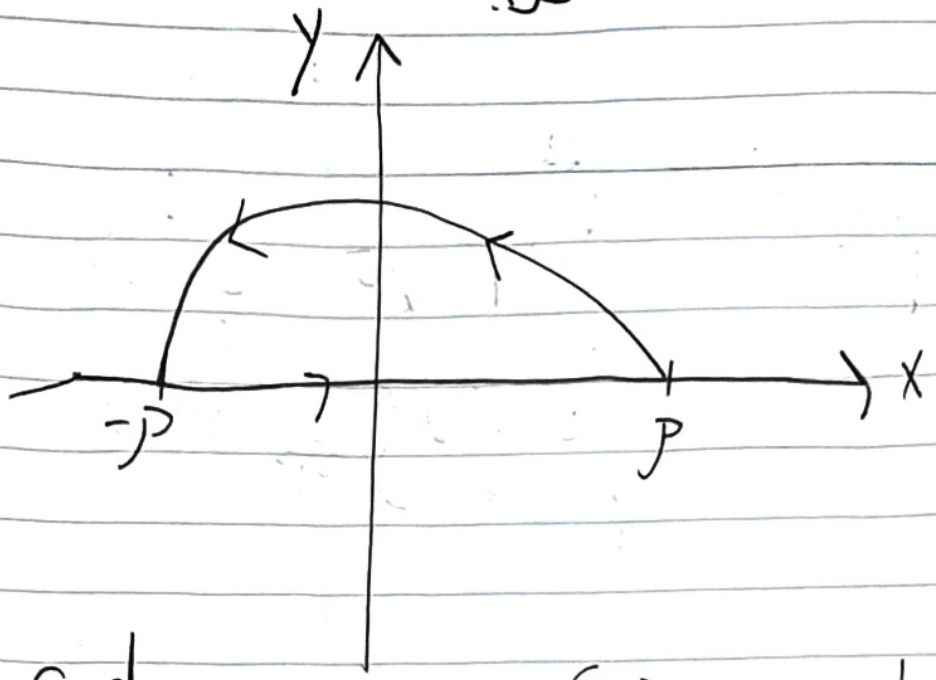
$r \to \infty$

drops

$$\oint_C \frac{\ln(z)}{1+z^2}\,dz = \int_0^\infty \frac{\ln(x)\,dx}{1+x^2} + \int_\infty^0 \frac{(\ln(x)+\pi i)\,dx}{1+x^2}$$

$$\oint_C \frac{\ln(z)}{1+z^2}\,dz = 2\int_0^\infty \frac{\ln(x)\,dx}{1+x^2} + \pi i \int_0^\infty \frac{dx}{1+x^2}$$

- $\int_0^\infty \frac{dx}{1+x^2} \rightarrow \frac{1}{2}\int_{-\infty}^{\infty} \frac{dx}{1+x^2}$



- $\oint \frac{dz}{1+z^2}$, $f(z) = \frac{1}{(z+i)(z-i)}$

- Poles: $z = i, -i$

- $R(i) = \lim_{z \to i} (z-i)\frac{1}{(z-i)(z+i)} = \frac{1}{2i}$

- $\oint \frac{dz}{1+z^2} = 2\pi i\left(\frac{1}{2i}\right) = \pi$

- $\oint \frac{dz}{1+z^2} = \int_{-p}^{p} \frac{dx}{1+x^2} + \int_0^\pi \frac{pie^{i\theta}\, d\theta}{1+p^2e^{2i\theta}}$ $p \to 0, \infty$

- $\oint \frac{dz}{1+z^2} = \int_{-\infty}^{\infty} \frac{dx}{1+x^2} = \pi$

- $\int_0^{\infty} \frac{dx}{1+x^2} = \frac{\pi}{2}$ $\quad I$

- $\oint \frac{\ln(z)}{1+z^2} dz = 2\int_0^{\infty} \frac{\ln(x)dx}{1+x^2} + \pi i \int_0^{\infty} \frac{dx}{1+x^2}$

- $\frac{\pi^2}{2} i = 2I + \pi i \left(\frac{\pi}{2}\right)$

- $\frac{\pi^2}{2} i - \frac{\pi^2}{2} i = 0 = 2I$

$I = 0$

$$\int_0^{\infty} \frac{\ln(x)dx}{1+x^2} = 0$$

Prob 3.)

$r \to$ is a constant

$$\int_0^\infty \frac{\sin(kr)}{k}\, dk$$

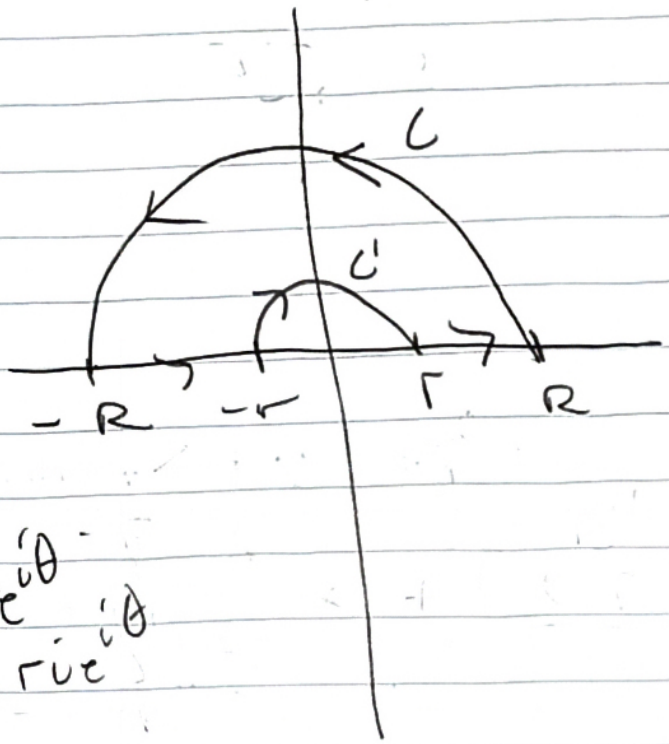
$\downarrow$

* $x = kr$
* $dx = r\,dk$

$$\frac{1}{r} \int_0^\infty \frac{\sin(x)}{x}\, dx$$

$$I = \int_0^\infty \frac{\sin(x)}{x}\, dx$$

$$z = x + iy = re^{i\theta}$$



$$\oint_{C'} \frac{e^{iz}}{z}\, dz \qquad z = re^{i\theta}$$

$$dz = rie^{i\theta}$$

$$\int_{C'} e^{iz}\, i\, d\theta$$

* $r \to 0$, $z \to 0$
* $e^{iz} \to 1$

$$\int_\pi^0 i\, d\theta = -i\pi$$

$R \to 0$, $r \to 0$

$$\int_{-\infty}^{\infty} \frac{e^{ix}}{x} dx - i\pi = 0$$

$$\int_{-\infty}^{\infty} \frac{e^{ix}}{x} dx = i\pi$$

$$Im\left(\int_{-\infty}^{\infty} \frac{e^{ix}}{x} dx \right) = Im(i\pi)$$

$$\int_{-\infty}^{\infty} \frac{e^{i\sin(x)}}{x} dx = \pi$$

$$\int_{0}^{\infty} \frac{\sin(x)}{x} dx = \frac{\pi}{2}$$

$$\int_{0}^{\infty} \frac{\sin(kr)}{k} dk = \frac{\pi}{2}$$

prob 4.)

- $\int_{-\infty}^{\infty} \frac{e^{-i\omega t}\, d\omega}{\omega^2 + a^2} = f(t)$

- $z = -i\omega$
- $dz = -i\, d\omega$

- $-\frac{1}{i}\int_{-\infty}^{\infty} \frac{e^{zt}\, dz}{-z^2 + a^2} = f(t)$

$z =$

- Bromwich integral:

$$f(t) = \frac{1}{2\pi i}\int_{-\infty}^{\infty} F(z)\, e^{zt}\, dz$$

$$* \quad \frac{f(t)}{2\pi} = \frac{1}{2\pi i}\int_{-\infty}^{\infty} \frac{e^{zt}\, dz}{z^2 - a^2}$$

- $f(t) =$ ~~Sum of Residues~~ $F(z)e^{zt}$ of all poles

- For fourier integrals at either use upper-semicircle or lower semicircle contour

(↳ upper semicircle)

* $F(z) = \frac{1}{z^2 - a^2} = \frac{1}{(z-a)(z+a)}$

* Poles : $z = a, \; z = -a$

* $R(a) = \cancel{(z-a)} \lim_{z \to a} \frac{1}{\cancel{(z-a)}(z+a)} = \frac{1}{2a}$

* $R(-a) = \cancel{(z+a)} \lim_{z \to -a} \frac{1}{(z-a)\cancel{(z+a)}} = \frac{-1}{2a}$

R: $F(z)e^{zt} = \frac{1}{2a} e^{at} - \frac{1}{2a} e^{-at}$

$\frac{f(t)}{2\pi} = \frac{e^{at}}{2a} - \frac{e^{-at}}{2a}$

If contour is upper semicircle

$$f(t) = \frac{\pi e^{-at}}{a}$$

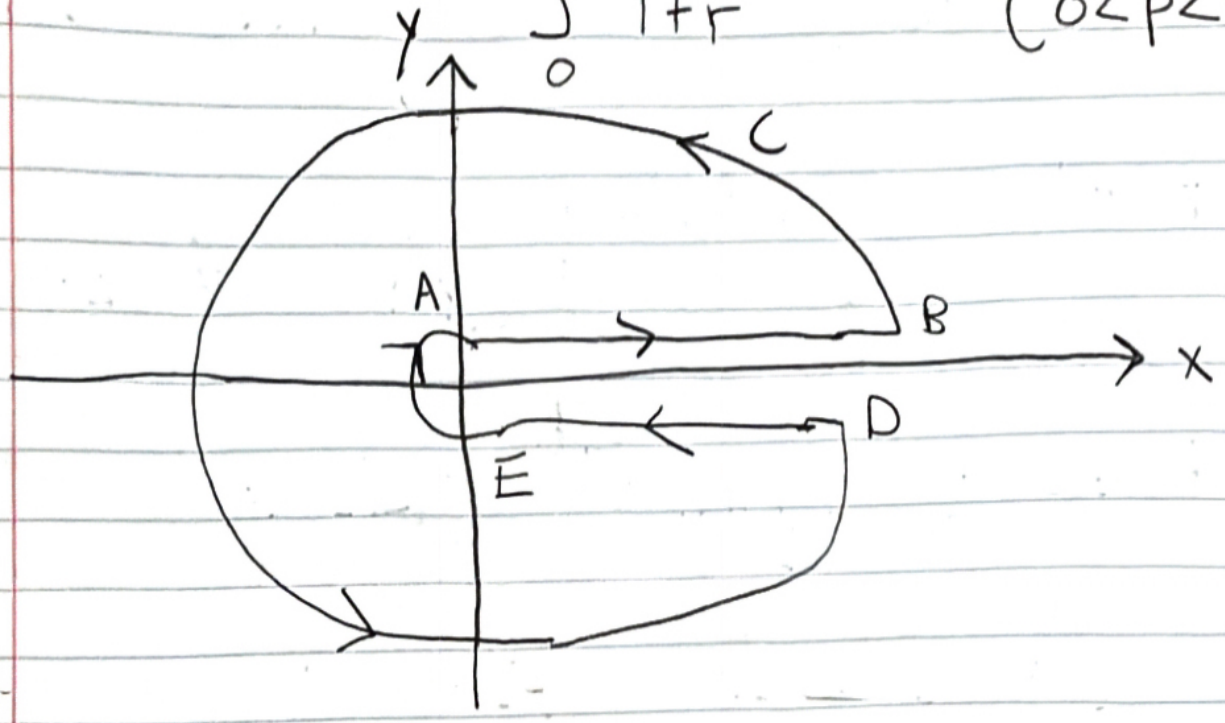
Prob 4)

* $|a|<1$

* $\int_0^\infty \frac{x^a}{(x+1)^2}\,dx$

Use a test example

Evaluate: * $\int_0^\infty \frac{r^{p-1}}{1+r}\,dr$ $(0<p<1)$



* $\oint \frac{z^{p-1}}{z+1}\,dz$, $0<p<1$

z^{p-1} , $z \rightarrow$ Multiple values of z^{p-1} (example of branches)

c/). $z = re^{i\pi/4}$ → $z^{-1/2} = r^{-1/2} e^{-i\pi/8}$

- θ increases 2π.
- $z = re^{i(\pi/4 + 2\pi)}$, $z^{-1/2} = r^{-1/2} e^{-i(\pi/8 + \pi)} = r^{-1/2} e^{-i(7\pi/8)}$

$$\oint \frac{z^{p-1}}{1+z} dz$$

Pole: $z = -1 = e^{i\pi}$

- $R(-1) = \lim_{z \to -1} \frac{(z+1) z^{p-1}}{(z+1)} = (-1)^{p-1}$
- $R(-1) = (e^{i\pi})^{p-1} = e^{i\pi p - i\pi} = -e^{i\pi p}$
- $\oint_C \frac{z^{p-1}}{1+z} dz = -2\pi i e^{i\pi p}$

for large and small circle regions:

- $\int \frac{r^{p-1} e^{i(p-1)\theta}}{1 + re^{i\theta}} ri e^{i\theta} d\theta = i \int \frac{r^p e^{ip\theta}}{1 + re^{i\theta}} d\theta$

$r \to 0$ & $r \to \infty$ tends 0

Segment A-B: $\theta = 0$

- $\int_0^\infty \frac{r^{p-1}}{1+r}\,dr$

Segment D-E: $\theta = 2\pi$

- $\int_\infty^0 \frac{r^{p-1}\, e^{2\pi i(p-1)}}{1 + re^{2\pi i}} e^{2\pi i}\,dr$

$= -\int_0^\infty \frac{r^{p-1} e^{2\pi i p}}{1+r}\,dr$

- Summing A-B & D-E:

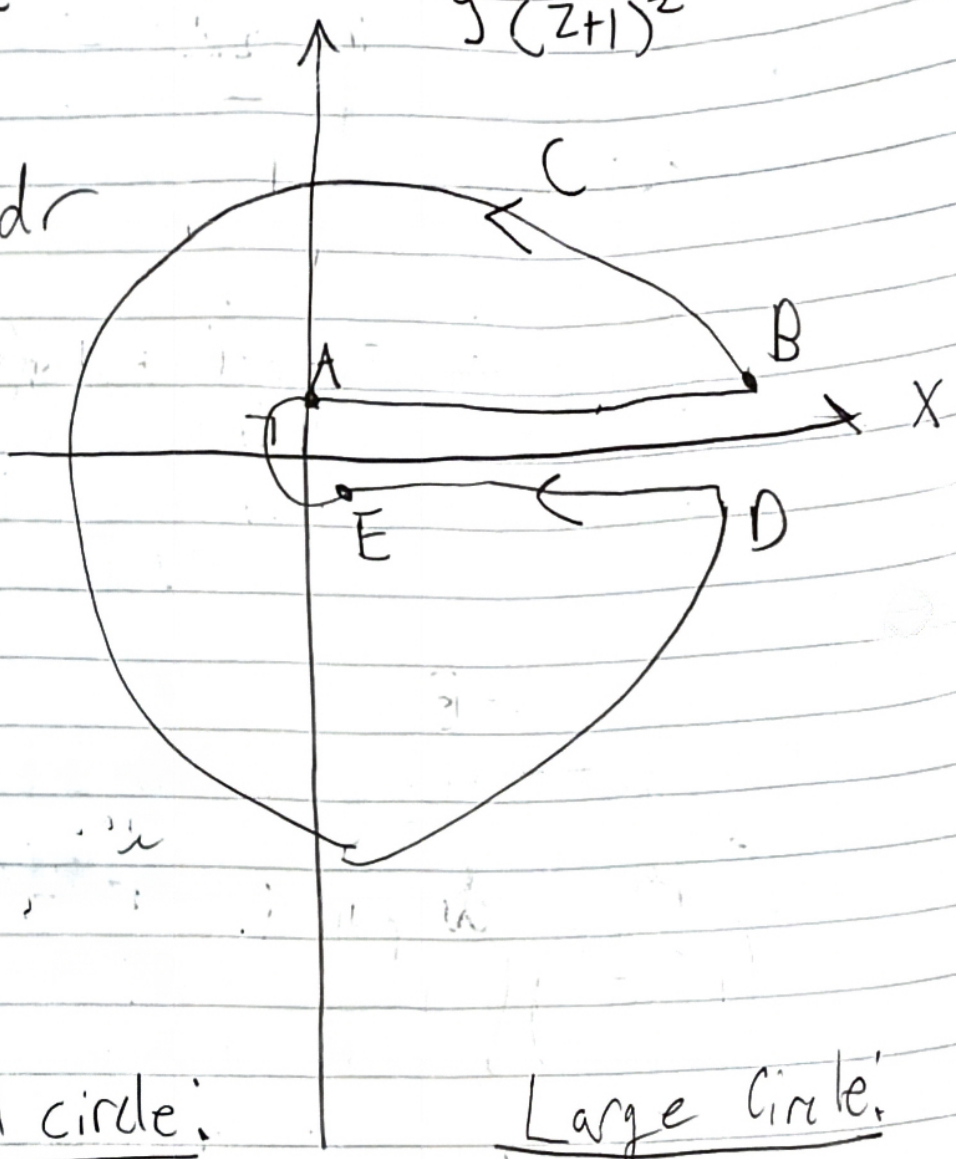
$$\left(1 - e^{2\pi i p}\right)\int_0^\infty \frac{r^{p-1}}{1+r}\,dr = -2\pi i e^{i\pi p}$$

$$\boxed{\int_0^\infty \frac{r^{p-1}}{1+r}\,dr = \frac{-2\pi i e^{i\pi p}}{1 - e^{2\pi i p}} = \frac{\pi}{\sin(\pi p)}}$$

$|a|<1$

$$* \int_0^\infty \frac{x^a}{(x+1)^2}\,dx \rightarrow \oint \frac{z^a}{(z+1)^2}\,dz$$

$$\int_0^\infty \frac{r^a}{(r+1)^2}\,dr$$



- $z = re^{i\theta}$
- $dz = rie^{i\theta}$

Small circle:

$$\int_0^{2\pi} \frac{r^a e^{i\theta a}}{(re^{i\theta}+1)^2} rie^{i\theta}\,d\theta$$

Large Circle:

$$\int_0^{2\pi} \frac{r^a e^{i\theta a}}{(re^{i\theta}+1)^2} rie^{i\theta}\,d\theta$$

Region A-B: $\theta = 0$

$$\int_0^\infty \frac{r^a}{(r+1)^2}\,dr$$

* $z = re^{2\pi i}$

* $dz = e^{2\pi i}\, dr$

Region D-E:

$$\int_{\infty}^{0} \frac{r^a e^{2\pi i a}\, e^{2\pi i}}{\left(1 + re^{2\pi i}\right)^2}\, dr \qquad \theta_i = 2\pi$$

$$\oint \frac{z^a}{(z+1)^2}\, dz = \text{Small circle} + \text{Large circle} + \text{D-E} + \text{A-B}$$

$r \to 0$, $r \to \infty$ $\Rightarrow 0$

D-E + A-B:

$$\int_0^{\infty} \frac{r^a}{(r+1)^2}\, dr + \int_{\infty}^{0} \frac{r^a e^{2\pi i a}}{(1+r)^2}\, dr$$

$$= \left(1 - e^{2\pi i a}\right) \int_0^{\infty} \frac{r^a}{(1+r)^2}\, dr$$

$$\oint \frac{z^a}{(z+1)^2}\, dz$$

Poles: $z = -1$

$$R(-1) = \frac{1}{1!} \frac{d}{dz}\left((z+1)^2 \frac{z^a}{(z+1)^2} \right)\Big|_{z=-1}$$

- $R(-1) = az^{a-1}\Big|_{z=-1} \qquad a(-1)^{a-1}$

- $R(-1) = a(-1)^{a-1} = ae^{i\pi(a-1)}$

  * $i\pi(a-1) = i\pi a - i\pi$

  * $e^{-i\pi} = -1$ *

- $R(-1) = ae^{i\pi(a-1)} \qquad e^{i\pi a} = -ae$

- $\oint \frac{z^a}{(z+1)^2}\,dz = -2\pi i a e^{i\pi a}$

- $\left(1 - e^{2\pi i a}\right)\int_0^\infty \frac{r^a}{(1+r)^2}\,dr = -2\pi i a e^{i\pi a}$

$$\int_0^\infty \frac{r^a}{(1+r)^2}\,dr = \frac{-2\pi i a e^{i\pi a}}{\left(1 - e^{2\pi i a}\right)}$$

* switch back variable *
say $x = r$

Prob 6.

a) * $g(z) = \pi \cot(\pi z)$

* Poles at $z = j\pi$ integer

* j is an integer $-\infty < j < \infty$.

* Find Residues.

* Period: $2\pi \curvearrowleft$

* $\cot(\pi z) = \dfrac{\cos(\pi z)}{\sin(\pi z)}$

$$\cos(\pi z) \approx 1 - \frac{(\pi z)^2}{2!} + \frac{(\pi z)^4}{4!} - \frac{(\pi z)^6}{6!}$$

$$\sin(\pi z) \approx (\pi z) - \frac{(\pi z)^3}{3!} + \frac{(\pi z)^5}{5!}$$

↑ divide 1-2nd terms

$$\cot(\pi z) \approx \frac{1 - \frac{\pi^2 z^2}{2!}}{\pi z - \frac{\pi^3 z^3}{3!}}$$

$$\pi z - \frac{\pi z^3}{6} \overline{\Big|\; \frac{\pi z}{2} \;\Big|\; \frac{-\pi^2 z^2}{2} + 1}$$

$$- \left(\frac{\pi z}{2} + \frac{\pi^2 z^4}{12} \right)$$

$$\frac{-\pi^2 z^4}{12} + 1$$

$$\cot(\pi z) \approx -\frac{\pi z}{2} + \frac{\left(\frac{-\pi^2 z^4}{12} + 1 \right)}{\left(\pi z - \frac{\pi z^3}{6} \right)}$$

* If you ignore the first terms of your sine and cosine series you'll notice that sine series has higher order $\cot(\pi z)$ is only evaluated as a simple pole.

$$R(j\pi) = \lim_{z \to j\pi} (z - j\pi)\pi \frac{\cos(\pi z)}{\sin(\pi z)}$$

$$l'h\left(\lim_{z \to j\pi} (z - j\pi)\pi \frac{\cos(\pi z)}{\sin(\pi z)} \right)$$

$$= \lim_{z \to j\pi} \pi \left(\frac{-\pi(z - j\pi)\sin(\pi z) + \cos(\pi z)}{-\pi\pi\cos(\pi z)} \right)$$

$$= \pi\left(\frac{\cos(j\pi^2)}{\pi\cos(j\pi^2)} \right) = 1$$

$$\boxed{R(j\pi) = 1}$$

b.) $S = \sum_{n=-\infty}^{\infty} f(n)$

Jordan's lemma:

- $\lim_{R \to \infty} f(z) = 0$ $\quad \left(\text{for all } z = Re^{i\theta} \right)$
- In the range $0 \leq \theta \leq \pi$
- $\lim_{R \to \infty} \int_C e^{iaz} f(z)\, dz = 0$

* ($a > 0$ C is a semicircle of radius R in the upper half-plane)

$$I = 2\pi i \sum_k \text{Res}\left(f(z)\pi \cot(\pi z), z_k \right)$$

$$I = \oint_C f(z)\pi \cot(\pi z)\, dz$$

* Contour is a circle *

- $\cot(x) = \frac{\cos(x)}{\sin(x)} = \left(\frac{e^{ix}+e^{-ix}}{2}\right)\left(\frac{2i}{e^{ix}-e^{-ix}}\right)$
- $\cot(x) = i\left(\frac{e^{ix}+e^{-ix}}{e^{ix}-e^{-ix}}\right)$
- $\cot(\pi z) = i\left(\frac{e^{i\pi z}+e^{-i\pi z}}{e^{i\pi z}-e^{-i\pi z}}\right)$
- lets say that $z = Re^{i\pi}$ (Large z) $|z| = |R| = m + \frac{1}{2}$
- $|\cot(\pi z)| = \left(\frac{|e^{i\pi z}| + |e^{-i\pi z}|}{|e^{i\pi z}| - |e^{i\pi z}|}\right)$

for large substitution:

$$\left|\cot(\pi z)\right| = \lim_{m\to\infty} \frac{e^{\pi(m+1/2)} + 1}{e^{\pi(m+1/2)} - 1} = 1$$

(for all Large numbers)

* $\left|\cot(\pi z)\right| \to 1$

if $m \to \infty$

* $\left|f(z)\, z\right| \to 0$

if $m \to \infty$

*

$$I = \pi \oint f(z)\cot(\pi z)\, dz$$

(Large z $(m \to \infty)$)

* $$\oint \left|f(z)\,dz\right| \leq \left|z\, f(z)\right| \to 0$$

(Large z)

So for $M \to \infty$ $I \to 0$

Part C.)

- $I = 2\pi i \sum_k \text{Res}\left(f(z)\pi \cot(\pi z), z_k \right)$
- as $M \to \infty$ from earlier residues tends to zero ($|zf(z)| \to 0$ for large $|z|$.)
- Residues theorem for $f(z)$:

- $I \to 2\pi i \sum_k \lim_{z \to z_k} (z - z_k) f(z) \pi \cot(\pi z)$

↑

(Any singularities of $f(z)$ term.)

Residue theorem for cot:

$\cot(\pi z)$ at $z = k$.

$$2\pi i \sum_{k=0}^{\infty} \operatorname{Res}\left(f(z)\,\pi\cot(\pi z), z_k\right)$$

$$= 2\pi i \sum_{n=0}^{\infty} f(n) \operatorname{Res}\left(\pi\cot(\pi z), n\right) \quad \left(k \to n \text{ switch}\right)$$

$$= 2\pi i \sum_{n=0}^{\infty} f(z)$$

$$I \to 2\pi i \left(\sum_{n=-\infty}^{\infty} f(n) + \sum_{k} \lim_{z \to z_k} (z - z_k) f(z) \pi \cot(\pi z) \right)$$

- Integral established earlier tends zero for large z.

$$S = \sum_{n=-\infty}^{\infty} f(n) = -\sum_{k} \lim_{z \to z_k} (z - z_k) f(z) \pi \cot(\pi z)$$

d.)

* $f(z) = \dfrac{1}{z^2 + a^2}$

$$z^2 + a^2 = 0$$

* $z = \pm ia$

$$R(ia) = \lim_{z \to ia} \left(\cancel{(z - ia)} \frac{1}{(z + ia)\cancel{(z - ia)}} \right) \pi \cot(\pi z)$$

$$R(ia) = \frac{\pi \cot(\pi ia)}{2ia}$$

$$R(-ia) = \lim_{z \to -ia} \left(\cancel{(z + ia)} \frac{1}{\cancel{(z + ia)}(z - ia)} \right) \pi \cot(\pi z)$$

$$R(-ia) = \frac{-\pi \cot(-\pi ia)}{2ia}$$

$$\sum_{n=-\infty}^{\infty} \frac{1}{n^2 + a^2} = \frac{\pi}{2ia} \left(-2 \cot(\pi ia) \right) = \frac{\pi}{ia} \cot(\pi ia)$$